

YUVAL FANG

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EDUCATION

University of California, San Diego

June 2027

Candidate for B.S. in Electrical Engineering

GPA: 3.84

Activities: Triton NeuroTech, Triton AI, Eta Kappa Nu (Engineering Honors Society), Sigma Phi Epsilon

Relevant Coursework: Pattern Recognition & ML, Linear and Nonlinear Optimization, Analog Design, Digital Design, Components & Circuits Lab, Linear Electronic Systems, Active Circuit Design, Programming in C, Java, C++

TECHNICAL EXPERIENCE

Hardware Engineer / Triton NeuroTech

September 2025 - Present

- Engineering the complete embedded system for an EMG/EEG-controlled wheelchair, managing hardware selection, assembly, and low-level firmware programming for movement.

Embedded Systems Engineer / Triton AI

September 2025 - Present

- Implementing a robust autonomous braking system by programming a PID control loop within an RTOS framework in C

Software Engineer Intern / Resolvem Inc.

July-September 2025

- Architected and deployed a serverless **business automation framework** on Vercel from Git using a modular **Node.js** (Express.js) backend, secured with a Microsoft OAuth2 authentication system.
- Developed a **calendar-to-task pipeline** that synchronizes Outlook events to Monday.com by orchestrating parallel batch requests to the Microsoft Graph and Monday.com GraphQL APIs, with robust error handling.
- Engineered an **invoice generation** module that securely accesses rate files from SharePoint via Microsoft Graph API with app-only authentication, dynamically **generating PDF reports** from task data received from Monday.com.
- Automated the end-to-end **purchase order** lifecycle with a module that processes receipt files, leveraging multer, exceljs and Claude+SharePoint API integration with RESTful endpoints for **document generation and record-keeping**.

Robot Assembly / Team Megatron

September-December 2024

- Engineered a **3D-printed robot chassis** and applied soldering/circuit integration to assemble ESP32-based motor control, sensors, and power systems for **autonomous navigation**.
- Implemented and calibrated **PID control algorithms** for line-following, iteratively optimizing design and performance. Team robot achieved **1st place** in speed competition out of 13 entrants.

Calculator Project

September-December 2024

- Designed a **bit receiving circuit** to receive Sony remote infrared signals. Programmed digital logic using Verilog which **read signals** using PWM, added inputs using a **four bit adder** and displayed on a **seven segment display**.
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VOLUNTEER EXPERIENCE

Lead Counselor / Tzofim, Orange County

09/2021 - 05/2024

- Developed and **led weekly educational activities** for counselors, contributing several hundred volunteer hours dedicated to mentorship and leadership building.
 - Planned and organized three annual **week-long camping trips** for 150 scouts.
 - Enhanced team-building skills by **supporting and problem-solving** with co-counselors.
 - Received 2x Gold President's Volunteer Service Award for outstanding service.
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SKILLS AND INTERESTS

- Technical Skills:** 3D Printing, Soldering, PSpice (OrCad), Breadboard Circuit Construction, Lab Tools
- Software Skills:** Proficient in Python, C, C++, MATLAB, Java, Node.js, Verilog, Microsoft Suite, Github
- Interests:** Surfing, Swimming. Bilingual. Clippers Fan. Radiohead.